

Universidad Rafael Landívar

Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y telecomunicaciones - sección 1

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REDES Y TELECOMUNICACIONES:

HOJA DE TRABAJO 9

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1.Pruebas de Gatewaey

PC1

```
Pinging 10.100.0.1 with 32 bytes of data:

Reply from 10.100.0.1: bytes=32 time<1ms TTL=255
Reply from 10.100.0.1: bytes=32 time<1ms TTL=255
Reply from 10.100.0.1: bytes=32 time<1ms TTL=255
Reply from 10.100.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.100.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

 Top

PC2

```
C:\>ping 10.100.32.1

Pinging 10.100.32.1 with 32 bytes of data:

Reply from 10.100.32.1: bytes=32 time<1ms TTL=255
Reply from 10.100.32.1: bytes=32 time<1ms TTL=255
Reply from 10.100.32.1: bytes=32 time=1ms TTL=255
Reply from 10.100.32.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.100.32.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

 Top

Rutas estáticas

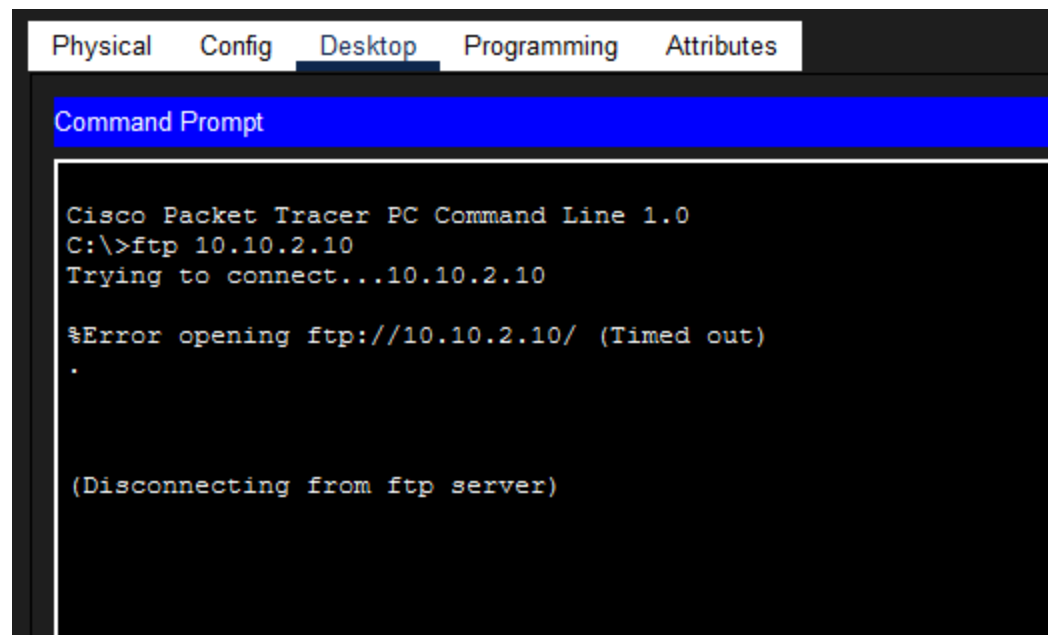
Gateway of last resort is not set

```
    172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.25.0/30 is directly connected, GigabitEthernet0/0
L       172.16.25.2/32 is directly connected, GigabitEthernet0/0
C       172.16.25.12/30 is directly connected, GigabitEthernet0/1
L       172.16.25.13/32 is directly connected, GigabitEthernet0/1
C       172.16.25.16/30 is directly connected, GigabitEthernet0/2
L       172.16.25.17/32 is directly connected, GigabitEthernet0/2
```

```
R1#show ip os
R1#show ip ospf ne
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.25.9	1	FULL/BDR	00:00:35	172.16.25.1	GigabitEthernet0/0

```
R1#
```



Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>

C:\>ping 10.10.0.10

Pinging 10.10.0.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 10.10.0.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.0.10

Pinging 10.10.0.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 10.10.0.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

4b

Segmento	Descripción	Red	Máscara (Bits/Decimal)	Primer Host Utilizable	Último Host Utilizable	Broadcast
VLAN 300	5000 Usuarios	10.100.32.10	/19 (255.255.224.0)	10.100.32.1	10.100.63.254	10.100.63.255
VLAN 200	500 Usuarios	10.100.0.10	/23 (255.255.254.0)	10.100.0.1	10.100.1.254	10.100.1.255
POD 01	Servidor WEB	10.10.0.10	/24 (255.255.255.0)	10.10.0.1	10.10.0.254	10.10.0.255
POD 03	Servidor Correo	10.10.1.10	/24 (255.255.255.0)	10.10.1.1	10.10.1.254	10.10.1.255
POD 04	Servidor FTP	10.10.2.10	/24 (255.255.255.0)	10.10.2.1	10.10.2.254	10.10.2.255
POD 05	Servidor DNS	10.10.3.10	/24 (255.255.255.0)	10.10.3.1	10.10.3.254	10.10.3.255

4c

Ítem (Router)	Red de Destino	CIDR	Máscara de Subred	Siguiente Salto (Next Hop)
R1 (Core)	POD 01 (Web)	/24	255.255.255.0	172.16.25.1 (Vía R2)
R1 (Core)	VLAN 200 y 300	/19	255.255.224.0	172.16.25.14 (Vía R5)
R2 (Dist.)	POD 01 (Web)	/24	255.255.255.0	172.16.25.18 (Vía R6)
R2 (Dist.)	POD 04 (FTP)	/24	255.255.255.0	172.16.25.22 (Vía R7)
R3 (Acceso)	VLAN 200 (500 Us.)	/23	255.255.254.0	Directamente Conectada
R3 (Acceso)	VLAN 300 (5k Us.)	/19	255.255.224.0	Directamente Conectada
R3 (Acceso)	Redes PODs / Core	/0	0.0.0.0 (Default)	172.16.25.25 (Vía R4)
R4 (Dist.)	POD 03 (Correo)	/24	255.255.255.0	10.10.1.1 (Int. Local)
R4 (Dist.)	Hacia R1 (Core)	/30	255.255.255.252	172.16.25.9 (Vía R1)
R5 (Dist.)	POD 05 (DNS)	/24	255.255.255.0	10.10.3.1 (Int. Local)
R5 (Dist.)	Hacia R3 (Acceso)	/30	255.255.255.252	172.16.25.30 (Vía R3)
R6 (POD 01)	Servidor WEB	/24	255.255.255.0	Directamente Conectada
R6 (POD 01)	Resto de la Red	/0	0.0.0.0 (Default)	172.16.25.17 (Vía R2)
R7 (POD 04)	Servidor FTP	/24	255.255.255.0	Directamente Conectada
R7 (POD 04)	Resto de la Red	/0	0.0.0.0 (Default)	172.16.25.21 (Vía R2)